Falcon 9–Class Reusable LOX/RP-1 Booster: Verified Parameter Dataset for an Engine-Cluster-Size Study

Bottom line: SpaceX itself publishes only a small set of numbers: per-engine sea-level thrust (845 kN), minimum throttle (108,300 lbf out of 190,000 lbf, which is 57%), liftoff mass (549,054 kg), expendable LEO payload (22,800 kg), upper-stage thrust and burn time (981 kN, 397 s), and launch-site coordinates. spacex Every Falcon 9 mass breakdown (dry mass, propellant split, landing propellant) and every specific impulse value is an unofficial estimate, including the widely used Dresia et al. (2021) Table A1. For a cluster-size study, the published 57% throttle floor is likely to be the deciding constraint, not engine dry mass.

TL;DR

- **Official SpaceX values are few.** The Falcon User's Guide (March 2025 edition) and spacex.com give 845 kN per Merlin 1D at sea level and a throttle range of 190,000 down to 108,300 lbf (57%). They also give 549,054 kg liftoff mass, 22,800 kg to LEO (expendable), and an upper stage of 981 kN for 397 s. All Falcon 9 dry mass, propellant and Isp values are unofficial. Dresia et al. (2021) list 27.2 t structural mass, 418.7 t propellant and 25.0 t landing propellant, and the authors call these "unofficial estimations."
- **Landing throttle, not engine mass, likely drives the answer.** At a fixed 7,605 kN total thrust, one engine at 57% gives landing thrust-to-weight (T/W) of about 1.6–1.8, depending on the landing mass assumed. Getting T/W ⩽ 1 at minimum throttle would need about 15 or more engines. Allowing T/W ⩽ 3 (a DLR feasibility threshold) needs about 5 or more. The classic linear Akin engine-mass formula adds only 59 kg per extra engine, so engine-mass penalties from clustering are second-order unless you use a power-law relationship.
- **Use published values as anchors, and calibrate the rest.** Generic mass formulas overpredict Merlin 1D mass by about 1.8× (Akin) to 2.7× (older log-law fit). Scale your formula to the Merlin data point (about 470 kg, secondary source) and run sensitivity cases over throttle floor (40–70%), landing propellant (15–35 t) and the engine-mass exponent.

Key Findings

1. **Thrust values in official sources don't fully agree.** The 2025 User's Guide text says 845 kN per engine and 7,605 kN total. Table 2-1 in the same document says 7,686 kN, spacex which is 9 × 854 kN, a figure carried over from the 2021 guide. mit mit 1,710,000 lbf converts to 7,606.5 kN. Use 845 kN and 7,605 kN, and treat 7,686 kN as a leftover from the older guide.
2. **Minimum throttle is officially 57% (Block 5, sea level).** mit spacex For the 2013 Merlin 1D, SpaceX gave a range of 70–100 percent, quoted by NASASpaceFlight in March 2013: "Added throttle capability for range from 70-100 percent". The 39–40% "deep throttle" claim has no primary source. The 2009 guide states the Merlin 1C (Falcon 9 v1.0) had no throttling. spaceflightnow
3. **Engine mass and T/W are not officially published.** SpaceX says only that Merlin has "the highest thrust-to-weight ratio of any boost engine ever made" (User's Guide). mit spacex The ">150" vacuum T/W claim dates from the 2013 qualification of the 147,000 lbf version (NASASpaceFlight, citing SpaceX). nasaspaceflight The 470 kg mass appears only in secondary sources. Wikipedia +2
4. **Dresia et al. 2021 Table A1 is confirmed as you cited it:** 27.2 t structural mass, 418.7 t propellant, 25.0 t landing propellant, 569.3 t GLOW, ascent Δv of 3.5 km/s and an assumed landing Δv of 2.0 km/s. The authors state the values "are based on unofficial estimations." dlr Their GLOW (569.3 t) is 20 t above SpaceX's 549.054 t, so their mass split doesn't add up to the official total. dlr spacex
5. **Published cluster-size results point in different directions and depend on the mass model.** Dresia's optimizer chose 5–6 first-stage engines, but the authors warn this is because their mass relationships "only consider the engine mass as a function of the engine thrust." dlr DLR's VTVL designs with fixed generic engines ended up with 15–17 hydrocarbon engines. dlr DLR notes that small stages can't land safely if T/W exceeds 3 even at 30% throttle, and that a European reusable engine would need 33–100% throttling to land with T/W near 1. dlr dlr

Source Key (cited by code in tables)

Code	Source	Type
S1	SpaceX, Falcon User's Guide, version 8 (change log: March 2025), file dated 2025-05-09. https://www.spacex.com/assets/media/falcon-users-guide-2025-05-09.pdf	Official
S2	SpaceX, Falcon User's Guide, September 2021 (version 7), copy hosted by MIT. https://web.mit.edu/2.70/Reading%20Materials/SpaceX%20Falcon-users-guide-2021-09.pdf	Official (third-party mirror)
S3	SpaceX, "Falcon 9" vehicle page, accessed Nov 2026. https://new.spacex.com/vehicles/falcon-9	Official
S4	SpaceX, Falcon 9 Launch Vehicle Payload User's Guide, 2009 (v1.0 era). https://www.spaceflightnow.com/falcon9/001/f9guide.pdf	Official (historical)
S5	Dresia, K., Jentzsch, S., Waxenegger-Wilfing, G., Dos Santos Hahn, R., Deeken, J., Oschwald, M., Mota, F., "Multidisciplinary Design Optimization of Reusable Launch Vehicles for Different Propellants and Objectives," J. Spacecraft and Rockets, 58(4), 2021, pp. 1017–1029, doi:10.2514/1.A34944. altmetric Open access: https://elib.dlr.de/141082/1/1.A34944.pdf	Peer-reviewed (its Falcon 9 data are unofficial estimates)
S6	Stappert, S., Wilken, J., Sippel, M., Dumont, E., "Assessment of a European Reusable VTVL Booster Stage," Space Propulsion 2018, SP2018_00478, Seville. https://elib.dlr.de/121410/2/478_STAPPERT.pdf	Conference (DLR)
S7	Akin, D. L., "Mass Estimating Relations," ENAE 791 lecture notes, Univ. of Maryland, 2016. https://spacecraft.ssl.umd.edu/academics/791S16/791S16L08.MERsx.pdf	Academic/secondary
S8	Zandbergen, B. T. C., "Simple mass and size estimation relationships of pump fed rocket engines for launch vehicle conceptual design," 6th EUCASS, 2015 (via ResearchGate)	Conference
S9	Zilker, F., "Aerothermal Analysis of Re-usable First Stage during Rocket Retro-propulsion," M.Sc. thesis, Univ. Stuttgart / DLR, 2018. https://elib.dlr.de/119265/1/SADA_ITLR.pdf	Academic thesis
S10	Blackmore, L., "Autonomous Precision Landing of Space Rockets," The Bridge, 46(4), NAE, 2016, pp. 15–20. http://www.larsblackmore.com/nae_bridge_2016.pdf	Peer-reviewed magazine (SpaceX author)
S11	Kyle, E. (Space Launch Report), "SpaceX Falcon 9 v1.2 Data Sheet," hosted by NASA Safety & Mission Assurance. https://sma.nasa.gov/LaunchVehicle/assets/spacex-falcon-9-v1.2-data-sheet.pdf	Secondary (on a government site)
S12	NASASpaceFlight, "Falcon 9 boost as Merlin 1D engine achieves major milestone," March 2013. https://www.nasaspaceflight.com/2013/03/falcon-9-boost-merlin-1d-engine-achieves-milestone/	Secondary
S13	Wikipedia, "SpaceX Merlin," "Falcon 9," "Falcon 9 Full Thrust" (accessed Nov 2026)	Unofficial
S14	shahar603, "Telemetry-Data" (webcast-extracted), SpaceX CRS-16 dataset, GitHub	Unofficial
S15	NASASpaceFlight forum thread quoting the SpaceX CRS-3 press kit (2014). https://forum.nasaspaceflight.com/index.php?topic=34464.0	Secondary (quotes official press kit)

Code	Source	Type
S16	SpaceX on X, 26 Jan 2023 (Starlink 5-2): “56 Starlink satellites—weighing in total more than 17.4 metric tons—marking the heaviest payload ever flown on Falcon”. Also reported by Space.com / Yahoo News, “SpaceX launches heaviest payload on reused rocket’s 9th flight,” Jan 2023, and by Spaceflight Now	Official (secondary reports agree)
S17	DLR SART: Sippel et al., IAF 2022 paper (https://elib.dlr.de/195201/1/IAF22-2.pdf); SLME system studies, Space Propulsion 2022 (https://elib.dlr.de/186560/1/SP-2022_Systematic_Studies_on_Reusable_Staged-Combustion_Rocket_Engine_SLME_for_European_Applications.pdf)	Conference (DLR)
S18	Engine data compiled by a research sub-agent from: ULA Delta II heritage page; Pratt & Whitney/NPO Energomash RD-180 fact sheet (via Spaceflight Now, 2002); Roscosmos 2012 reproduction of Energomash tables; SpaceX post on X, 3 Aug 2024 (Raptor 1/2/3); Rocket Lab Electron page; Metal AM, May 2026; NASASpaceFlight, Nov 2016 (YF-100); Blue Origin fact sheet via Spaceflight Now, 2014; enginehistory.org; Purdue AAE; Astronautix. Search snippets only, full pages not opened.	Mixed (marked per row)
S19	Wevolver spec pages (aggregator of uncertain origin)	Unofficial
S20	Marwege, A. et al., “RETALT: review of technologies and overview of design changes,” CEAS Space Journal, 2022. https://pmc.ncbi.nlm.nih.gov/articles/PMC9244295/	Peer-reviewed

Confidence scale: **High** = official or primary, checked in the full document. **Medium** = peer-reviewed or reputable secondary source, or official but internally inconsistent. **Low** = unofficial, single source, or snippet only.

A. Engine: Merlin 1D (sea level) and earlier versions

Parameter	Value	Units	Version	Source	Page/Table	Source type	Confidence	Notes
Sea-level thrust, per engine	845 (190,000 lbf) spacex	kN	Block 5 / current	S1; S3	S1 §2.3	Official	High	Current official value. spacex
Sea-level thrust, per engine	854 (190,000 lbf) mit	kN	FT/Block 5 (2021)	S2	§2.3	Official	Medium	190,000 lbf is actually 845 kN, so 854 is likely a typo. mit
Sea-level thrust, 9-engine total	7,605 (text) / 7,686 (Table 2-1)	kN	Block 5	S1	§2.3; Table 2-1	Official	Medium	Internal mismatch. 7,686 = 9 × 854. Use 7,605. spacex
Sea-level thrust, 9-engine total	7,607 Wikipedia	kN	FT	S5; S13	S5 Table A1	Peer-reviewed / unofficial	Medium	Matches 1,710,000 lbf. Wikipedia
Vacuum thrust, 9-engine total	8,227 (914 per engine) dlr Wikipedia	kN	FT	S5; S13	S5 Table A1	Peer-reviewed (unofficial data)	Medium	No current official vacuum value for the sea-level Merlin. Wikipedia
Sea-level thrust, per engine	556 (125,000 lbf) Wikipedia	kN	Merlin 1C (v1.0)	S4	—	Official (historical)	High	v1.0 had 9 engines in a 3×3 grid. Wikipedia spaceflightnow Wikipedia
Stage sea-level / vacuum thrust	5,885 / 6,672 Wikipedia	kN	v1.1	S13	—	Unofficial	Low	v1.1 used early Merlin 1D, about 147,000 lbf per engine. nasaspaceflight Wikipedia
Sea-level / vacuum Isp	282 / 311 Wikipedia	s	v1.1 era, still widely quoted	S13	—	Unofficial	Low	Wikipedia marks these “needs update.” Wikipedia
Sea-level / vacuum Isp	283 / 312 dlr	s	FT/Block 5	S5	Table A1	Peer-reviewed (unofficial data)	Medium	SpaceX publishes no Isp for the sea-level Merlin.

Parameter	Value	Units	Version	Source	Page/Table	Source type	Confidence	Notes
Dry mass	~470 Wikipedia	kg	Merlin 1D (version unclear)	S13	—	Unofficial	Low	No primary source found. Wikipedia and Wikidata (Q19923939) both list 470 kg with no SpaceX citation. Astronautix gives 490 kg for an early version.
Thrust-to-weight	>150 (vacuum) nasaspaceflight	–	Merlin 1D, 147 klbf version (2013)	S12 (citing SpaceX)	—	Secondary	Medium	NASASpaceFlight, March 2013: "The Merlin 1D has a vacuum thrust-to-weight ratio exceeding 150, the best of any liquid rocket engine in history." The 28-test qualification program ran 1,970 s, including four tests at or above 147,000 lbf for 185 s.
Thrust-to-weight	"highest of any boost engine ever made" (no number) mit	–	Block 5	S1	§2.3	Official	High	Qualitative only. spacex
Thrust-to-weight	~184 (845 kN ÷ (470 kg × 9.81)) Wikipedia	–	Block 5	S13; derived	—	Unofficial / derived	Low	Only as good as the 470 kg mass. Wikipedia
Minimum throttle	108,300 of 190,000 lbf = **57%** mit	%	Block 5 (sea level)	S1; S2	Table 2-1	Official	High	Best value for this study. mit spacex
Minimum throttle	70% (range 70–100 percent)	%	Early Merlin 1D (2013)	S12 (quoting SpaceX)	—	Secondary (quotes SpaceX)	Medium	SpaceX statement quoted by NASASpaceFlight (Chris Bergin, March 2013): "Added throttle capability for range from 70-100 percent."

Parameter	Value	Units	Version	Source	Page/Table	Source type	Confidence	Notes
Minimum throttle	~39–40%	%	Claimed for Merlin 1D	Forums only	—	Unofficial	Low	Unverified. Use only as a sensitivity case. nasaspaceflight spacelaunchlive
Throttle	None	-	Merlin 1C (v1.0)	S4 (via S18)	—	Official (historical)	Medium	spaceflightnow
Chamber pressure	9.7 (1,410 psi)	MPa	Merlin 1D	S13	—	Unofficial	Low	Other unofficial values: 10.8 MPa (S19, "1D++"); 9.6 MPa (Astronautix, MVac). astronautix Wikipedia +2
Mixture ratio (O/F)	2.36	-	"Merlin-1D++"	S19	—	Unofficial	Low	Consistent with RP-1 gas-generator practice. wevolver
Nozzle expansion ratio	16	-	Merlin 1D	S19	—	Unofficial	Low	DLR used ε = 20 for generic reusable gas-generator engines (S6). dlr wevolver
Throat / exit area	0.042 / 0.90	m^2	Merlin 1D	S19	—	Unofficial	Low	0.90 m^2 means about 1.07 m exit diameter, which conflicts with the 0.92 m engine diameter on Wikipedia. Wikipedia wevolver
Propellant mass flow	~305 (845 kN ÷ (282 s × g_0))	kg/s	Block 5	Derived	—	Engineering estimate	Low	Not published. Calculated from thrust and Isp.
Engine cycle	Gas generator; single-shaft turbopump; pintle injector; regeneratively cooled copper-alloy liner mit	-	Merlin 1D	S1	§2.3	Official	High	Steering hydraulics use RP-1 as the working fluid. mit spacex
Restart	Yes mit	-	Merlin 1D	S1	Table 2-1	Official	High	spacex

Parameter	Value	Units	Version	Source	Page/Table	Source type	Confidence	Notes
Boostback engines	3 dlr nasa	count	FT (when flown)	S9; S11	S9 §1.2	Thesis / secondary	Medium	Drone-ship missions often skip boostback or do a partial one.
Entry burn engines	3 (about 20 s) nasa	count, s	FT/v1.2	S9; S11	—	Thesis / secondary	Medium	
Landing burn engines	1 (center engine, about 32 s) usually; a 3-engine landing burn has been tried nasa	count, s	FT/v1.2	S11; S9	—	Secondary	Medium	32 s is from one flight entry in S11. Wikipedia

B. First stage / booster

Parameter	Value	Units	Version	Source	Page/Table	Source type	Confidence	Notes
Liftoff mass (full vehicle)	549,054 spacex	kg	Current (FT/Block 5)	S3	Overview	Official	High	
Liftoff mass	569.3 dlr	t	“v1.2 Block 5”	S5	Table A1	Peer-reviewed (unofficial data)	Medium	20 t more than official.
Liftoff mass	505,846 / 333,400 tesla	kg	v1.1 / v1.0	S13	—	Unofficial	Low	
First-stage structural (dry) mass	27.2 dlr	t	Block 5	S5	Table A1	Peer-reviewed (unofficial data)	Medium	Confirmed as you cited it.
First-stage “empty mass”	25,600 wevolver	kg	Block 5	S19	—	Unofficial	Low	Origin unknown. Community estimates run about 22–27 t.
First-stage propellant	418.7 dlr	t	Block 5	S5	Table A1	Peer-reviewed (unofficial data)	Medium	
Landing propellant (drone ship)	25.0 dlr	t	Block 5	S5	Table A1	Peer-reviewed (unofficial data)	Medium	From reverse engineering. Optimizer gave 26.6 t. dlr
Landing Δv	2,000 (includes 5% margin), for 3,500 ascent Δv dlr	m/s	Block 5	S5	§II.B.2	Peer-reviewed	Medium	A modeling assumption.
Descent propellant share	7–9% of stage GLOM (hydrocarbon DLR stages) dlr	%	DLR concepts	S6	§4	Conference	Medium	For drone-ship landing.
Diameter	3.66 (SpaceX also rounds to 3.7) mit	m	All	S1; S3	Table 2-1	Official	High	spacex

Parameter	Value	Units	Version	Source	Page/Table	Source type	Confidence	Notes
First-stage length	41.2 (47.7 with interstage) / 40.9 dlr wevolver	m	FT/Block 5	S19; S13 / S5	Table A1	Unofficial / peer-reviewed	Low–Medium	Not on the current official page.
Recovery hardware	4 grid fins, 4 deployable legs mit	count	FT/Block 5	S1; S3	§2.3	Official	High	No masses published. spacex
Legs + grid fins mass	Not publicly available. Proxy: +15% of first-stage dry mass dlr	%	Model	S5	§II.A	Peer-reviewed	Low	DLR (S6) scales leg and fin mass linearly with dry mass and calls this conservative. dlr
Engine share of dry mass	~16% (9 × 470 kg ÷ 27.2 t)	%	Block 5	Derived from S13, S5	—	Engineering estimate	Low	Dresia's optimized RP-1 first stages have propulsion (engines + steering) at about 28% of structural mass. dlr
Expendable LEO payload	22,800 spacex	kg	Current	S3	Overview	Official	High	Target orbit not stated by SpaceX. Wikipedia
Reusable LEO payload (demonstrated)	>17,400 (56 Starlink, Starlink 5-2, 26 Jan 2023; booster B1067 on its 9th flight, landed on drone ship JRTI)	kg	Block 5	S16 (SpaceX on X)	—	Official	High	SpaceX: "56 Starlink satellites—weighing in total more than 17.4 metric tons—marking the heaviest payload ever flown on Falcon." Spaceflight Now (Stephen

Parameter	Value	Units	Version	Source	Page/Table	Source type	Confidence	Notes
								Clark) reported “more than 38,000 pounds.” About 24% below expendable.
Reusable LEO payload (quoted capability)	17,500 / 18,500 Wikipedia Wikipedia	kg	FT/Block 5	S13	—	Unofficial	Low	Wikipedia pages disagree with each other. Wikipedia Wikipedia
GTO payload, expendable / drone ship	8,300 / ~5,500 Wikipedia spacex	kg	FT/Block 5	S3 / S6; S13	—	Official / conference	High / Medium	About a 34% penalty for reuse. Wikipedia
MECO (v1.1, CRS-3)	t ≈ 161 s, h ≈ 80 km, about Mach 10 nasaspaceflight	s, km	v1.1	S15	—	Secondary (quotes press kit)	Medium	
MECO (Block 5, CRS-16, RTLS)	145 s; 1,624 m/s; 66.9 km; peak 3.43 g github	s, m/s, km	Block 5	S14	—	Unofficial	Low	A return-to-launch-site mission, so MECO is earlier and slower than drone-ship missions.
MECO for LEO drone-ship missions; flight-path angle	Not publicly available	–	–	–	–	–	–	Proxy: extract from webcast telemetry for a specific drone-ship mission (S14 method).
Landing distance downrange	~300 (CRS-8) Wikipedia	km	FT	S13	—	Unofficial	Low	Varies by mission.

Parameter	Value	Units	Version	Source	Page/Table	Source type	Confidence	Notes
Entry burn start altitude	~70–50 (F9 about 50) dlr dlr	km	FT	S9; S6	S6 §4	Thesis / conference	Medium	DLR's reconstruction of the SES-10 flight.
Touchdown velocity	Not publicly available	m/s	–	–	–	–	–	Proxy: about 2 m/s or less (engineering assumption).
Reentry q / heating limits (model)	q < 200 kPa; lateral load < 3 g; heat flux capped at the SES-10 reentry value dlr	kPa, g	DLR model	S6	§3.4	Conference	Medium	These are model limits, not SpaceX data.
Retro-propulsion heating	Side wall up to ~20 kW/m²; base up to ~70 kW/m²; side wall up to ~400 K (NASA infrared imaging showed about 450 K) dlr	kW/m², K	DLR 9-engine concept	S9	Abstract	Thesis	Medium	Entry burn from Mach 9.45 to 5.09 between 70 and 36 km. dlr
Drag coefficient for descent	Not given numerically in the sources reviewed. Dresia assumes a constant Cd and uses √(m/A) ≈ 1.47–1.62 t^½/m for RP-1 stages dlr	–	Model	S5	Table 1, Eq. 4	Peer-reviewed	Low	RETALT1's aerodynamic database (S20) is open access and is the best source for Mach-dependent Cd. nih
Precision landing approach	Boostback targeting, grid-fin control in the atmosphere, landing burn; 99% of dispersions must fall inside the landing zone nationalacademies larsblackmore	–	F9R	S10	pp. 15–20	Peer-reviewed (SpaceX author)	Medium	Qualitative only, no numbers.

C. Upper stage (Merlin Vacuum)

Parameter	Value	Units	Version	Source	Page/Table	Source type	Confidence	Notes
Vacuum thrust	981 (220,500 lbf) spacex mit	kN	Block 5	S1; S3	Table 2-1	Official	High	spacex
Vacuum thrust	934 / 801 / 617 Wikipedia	kN	FT / v1.1 / v1.0	S13	—	Unofficial	Low	astronautix Wikipedia
Minimum throttle	140,679 of 220,500 lbf ≈ 64% mit	%	Block 5	S1	Table 2-1	Official	High	spacex
Vacuum Isp	348 tesla Wikipedia	s	FT/Block 5	S5; S13	Table A1	Peer-reviewed (unofficial data)	Medium	Not official. Wevolver also lists 348 s. wevolver Wikipedia
Expansion ratio	165:1, fixed nozzle mit	–	Block 5	S1	§2.3	Official	High	spacex
Burn time	397 spacex	s	Block 5	S3	Second stage	Official	High	
Dry mass	4.5 dlr	t	Block 5	S5	Table A1	Peer-reviewed (unofficial data)	Medium	
Propellant	111.5 dlr	t	Block 5	S5	Table A1	Peer-reviewed (unofficial data)	Medium	Older secondary figure: 92,670 kg (Leitenberger, v1.1 era). bernd-leitenberger
Fairing dimensions	5.2 diameter × 13.2 height (extended version 18.7) mit	m	Current	S1	§4.1	Official	High	Composite: aluminum honeycomb with carbon-fiber skins (S2). mit spacex
Fairing mass; payload adapter mass	Not publicly available	kg	–	–	–	–	–	Proxy: Dresia's "payload bay" of 7.4 t for a 5.0 t payload implies about 2.4 t for fairing, adapter and avionics bay (S5), or use Akin's fairing formula m = 4.95·A^1.15 (S7). dlr umd

Parameter	Value	Units	Version	Source	Page/Table	Source type	Confidence	Notes
Adapter capacity	1,575 mm: up to 10,885 kg; 2,624 mm: up to 19,050 kg; 3,117 mm strut: up to 26,500 kg	kg	Current	S1	Table 4-1	Official	High	Structural limits, not performance. spacex

D. Mission / constraints

Parameter	Value	Units	Version	Source	Page/Table	Source type	Confidence	Notes
Maximum dynamic pressure (max-Q)	Not published (acronym only)	kPa	–	S1	Acronyms	Official	–	Proxy: about 30–35 kPa (community estimates, unofficial). nasaspaceflight +3
Payload design load factors	Axial up to +6.0 g, −2 g (payloads > 1,810 kg); up to +8.5 g (lighter payloads) mit	g	2021	S2	Figs. 4-1, 4-2	Official	High	Limit loads including dynamics, not the steady-state acceleration cap.
Steady-state acceleration control	“Both the first- and second-stage engines may be throttled to help maintain... steady state acceleration limits” (no number) mit	–	2021	S2	§4.3.1	Official	High	The 2009 guide shows shutting down 2 engines for the acceleration limit (v1.0). spaceflightnow
SLC-40 latitude	28.5620° N, 80.5772° W mit	deg	–	S2	§6.1.1	Official	High	
LC-39A latitude	28.6082° N, 80.6041° W mit	deg	–	S2	§6.1.2	Official	High	
LEO inclinations, Eastern Range	28.5–55 (55–65 with a dog-leg penalty)	deg	2025	S1	Table 3-1	Official	High	Below 28.5° costs performance. mit spacex
Mass-to-orbit curves	“available upon request” mit	–	2021/2025	S1; S2	§3.2	Official	High	No official performance curves are published. spacex
Orbit for 22,800 kg figure	Not stated by SpaceX. Secondary sources: 200 km LEO from Cape	km, deg	FT	Leitenberger (secondary)	—	Secondary	Low	Fits your provisional 200 km target.

Parameter	Value	Units	Version	Source	Page/Table	Source type	Confidence	Notes
	(~28.5°) bernd-leitenberger							
Dresia LEO mission	200 × 200 km; Δv_orbit 8,030 m/s; Δv with losses 9,500 m/s dlr	m/s	Model	S5	Table 2	Peer-reviewed	Medium	Losses assumed: gravity 1,000–1,500, drag 100–150 m/s. dlr

E. Engine mass scaling data and mass estimating relationships (MERs)

E1. Engine table (T/W = thrust ÷ (mass × 9.81) unless the source gives it)

Engine	Sea-level / vacuum thrust (kN)	Dry mass (kg)	T/W	Cycle / propellant	Source	Source type	Confidence	Notes
Merlin 1D (Block 5)	845 / ~914	~470	~184 (SL)	Gas generator / LOX-RP-1	S1 (thrust); S13 (mass)	Official / unofficial	Medium / Low	Mass is undated. Wikipedia spacex
Merlin 1C (F9 v1.0)	556 / — Wikipedia	Not found	—	Gas generator / LOX-RP-1	S4	Official (thrust)	Medium	spaceflightnow
F-1	6,770 / 7,770	8,353–8,400	~94	Gas generator / LOX-RP-1	NASA MSFC (thrust); Wikipedia/Astronautix (mass)	Government / secondary	Medium	Uprated to 1,522,000 lbf from Apollo 8 onward. Wikipedia +2
H-1 (Saturn IB)	912 / —	635 / 911 / ~1,000 (conflicting)	~93–146	Gas generator / LOX-RP-1	enginehistory.org (citing Rocketdyne manual); Purdue	Secondary	Low	The lowest mass may leave out some hardware. Wikipedia +2
RS-27A	890 / 1,054	1,147	~79 (SL)	Gas generator / LOX-RP-1	ULA Delta II page	Official	High	Wikipedia ulalaunch
RD-180	3,827 / 4,152	5,393–5,480	~72	Oxygen-rich staged combustion / LOX-RP-1	P&W/Energomash fact sheet; Roscosmos	Official	High	Throttle 40–100%. Wikipedia spaceflightnow
RD-170 / RD-171M	7,250 / 7,900	9,750	~76–82	Oxygen-rich staged combustion / LOX-RP-1, 4 chambers	Energomash via Roscosmos	Near-official	Medium	Wikipedia roscosmos
RD-107A	839.5 / 1,019.9	1,190 (RD-107)	~72	Gas generator / LOX-kerosene, 4 chambers	Wikipedia	Unofficial	Low	No RD-108A data found. Wikipedia
NK-33	1,510 / 1,680	1,240	~124	Oxygen-rich staged combustion	Wikipedia	Unofficial	Low	Wikipedia

Engine	Sea-level / vacuum thrust (kN)	Dry mass (kg)	T/W	Cycle / propellant	Source	Source type	Confidence	Notes
				/ LOX-kerosene				
RD-191	1,922 / 2,085	2,200 (official) / 2,290	~89	Oxygen-rich staged combustion / LOX-kerosene	Energomash via Roscosmos	Near-official	Medium	RD-191M throttle 27–110% (secondary). Wikipedia +2
YF-100	1,199 / 1,339	Not found	—	Oxygen-rich staged combustion / LOX-kerosene	NASASpaceFlight 2016	Secondary	Low	nasaspaceflight
Rutherford (small-engine end)	~24.9 / 25.8	35	~73	Electric pump / LOX-RP-1 Wikipedia	Rocket Lab; Metal AM 2026	Official / trade press	Medium	Wikipedia metal-am
Methane: Raptor 2	2,256 / —	1,630	~141	Full-flow staged combustion / LOX-CH4	SpaceX on X, 2024	Official	High	newspaceeconomy Wikipedia
Methane: Raptor 3	2,746 / —	1,525 (engine only)	~184	Full-flow staged combustion / LOX-CH4	SpaceX on X, 2024	Official	High	1,720 kg including vehicle-side hardware. x
Methane: BE-4	2,450 (550,000 lbf baseline) / —	Not found	—	Oxygen-rich staged combustion / LOX-LNG	Blue Origin via Spaceflight Now 2014; ULA BE-4 fact sheet; NASASpaceFlight, Nov 2025	Official (thrust) / secondary (uprate)	Medium	ULA fact sheet confirms “550,000-lbf thrust at sea level.” In Nov 2025, NASASpaceFlight reported test-stand engines at “625,000 pounds-force under current conditions, with plans to reach 640,000 pounds-force… through propellant

Engine	Sea-level / vacuum thrust (kN)	Dry mass (kg)	T/W	Cycle / propellant	Source	Source type	Confidence	Notes
								subcooling." That takes the 7-engine New Glenn booster from 17,219 kN to 19,928 kN.

The Merlin 1D is an outlier. Its T/W of about 184 is 2–2.5× that of other gas-generator kerosene engines (RS-27A about 79, F-1 about 94). Any MER fitted to historical engines will overpredict Merlin mass badly unless you calibrate it.

E2. Published MERs

MER	Functional form	Validity range	Source	Source type	Confidence	Check against Merlin 1D (845 kN, ε = 16)
Akin (UMd), liquid pump-fed engine	$m\ (kg) = 7.81\times10^{-4}\cdot T(N) + 3.37\times10^{-5}\cdot T(N)\cdot\sqrt{(Ae/At)} + 59$	Not stated (based on NASA CR-2420, TM-78661, TN-D-6349, JSC-26098) umd	S7, slide 25	Academic	Medium	≈ 833 kg, which is 1.8× the 470 kg figure
Akin, thrust structure	$m\ (kg) = 2.55\times10^{-4}\cdot T(N)$ umd	Not stated	S7, slide 25	Academic	Medium	7,605 kN total → 1.94 t, the same for any engine count
Akin, gimbals	$m\ (kg) = 237.8\cdot(T/p_0)\text{^}0.9375$ umd	Not stated	S7, slide 26	Academic	Medium	Depends on chamber pressure p_0
Zandbergen M-C2 (hydrogen-oxygen)	$m = 1.866\times10^{-10}\cdot T^2 + 0.00130\cdot T + 77.4$ (T in N)	15 kN–8 MN; researchgate typical error about 13%	S8, Table 4	Conference	Medium	Hydrogen-oxygen only. Do not use for RP-1. researchgate researchgate
Zandbergen, kerosene and storable	Multi-parameter form (thrust, number of chambers, expansion ratio). The snippet shows "M = (1.079E-3·THRUST + 53.0165)·NT^0.0369·…"	15 kN–8 MN; stated accuracy ±10–25% researchgate	S8, Table 4	Conference	Low	Full coefficients not verified. Get the paper.
Older log law (cited as ref. [5] in Zandbergen)	$T/W = 25.2\cdot\log_{10}(F[N]) - 80.7$	Small dataset (8 engines)	S8	Conference (secondary)	Low	T/W ≈ 69, so mass ≈ 1,250 kg (2.7×) researchgate
Castellini (2012), used by Dresia	Engine mass as a function of thrust only dlr	—	S5 (citing Castellini, PhD thesis, Politecnico di Milano)	Peer-reviewed (indirect)	Low	Exact form not retrieved
Rohrschneider (2002), Georgia Tech MER database	Not retrieved	—	Georgia Tech SSDL M.S. project	Academic	—	Exponents not verified semanticscholar

MER	Functional form	Validity range	Source	Source type	Confidence	Check against Merlin 1D (845 kN, ε = 16)
Tizón & Román, AIAA 2017-5010	Not retrieved	—	—	—	—	Not verified

What the MERs mean for engine count at fixed total thrust: With Akin's linear form, total engine mass is 7.02 t + 0.059 t × N for 7,605 kN at ε = 16. That gives about 7.08 t for N = 1, 7.55 t for N = 9 and 8.61 t for N = 27. Clustering costs only about 59 kg per engine, so engine dry mass alone won't decide the optimum. A power-law MER with exponent below 1 would make the clustering penalty larger. The hydrogen-oxygen quadratic does the opposite: it penalizes big engines. So the result depends on which formula you choose, and you should report it across a range of exponents.

E3. Thrust-structure and multi-engine relationships

- Akin: thrust structure scales only with total thrust (2.55×10^{-4} kg/N) and does not depend on engine count (S7). umd
- DLR uses empirical formulas for thrust frame, engine equipment, steering, controllers and harness inside its stsm tool. dlr Their exact form isn't published in S6 (S6 §3.3).
- No published relationship that ties thrust-structure mass explicitly to engine count was found. **Not publicly available.** A defensible proxy: Akin's formula plus a per-engine add-on for steering, plumbing and controller hardware, varied in sensitivity runs (engineering assumption).

F. Published studies on engine count for reusable vertical-landing boosters

Study	What was varied	Finding	Source type	Confidence	Notes
Dresia et al., JSR 2021 (S5)	Engine count 5–15 (first stage), chamber pressure, mixture ratio, expansion ratio, Δv split; minimum acceleration 1.3 g	Optimal RP-1 first stages had **5–6 engines**. The authors say the optimum "seems to be relatively small" but that this follows from mass correlations that "only consider the engine mass as a function of the engine thrust." dlr	Peer-reviewed	Medium	Doesn't model a landing throttle limit explicitly. Assumes constant landing Δv. dlr
Stappert et al., Space Propulsion 2018 (S6)	Propellants, staging, engine cycle; DLR's generic reusable gas-generator engines	Hydrocarbon VTVL launchers for 7.5 t to GTO needed **15–17 engines**, which only fit with slight overhang. Hydrogen versions needed 9. dlr	Conference	Medium	Engine count came from liftoff T/W, not optimization. Deep-throttle transients were not modeled. dlr
DLR SART, IAF 2022 and SLME studies (S17)	Small and large reusable stages	For small stages, landing T/W **"would exceed 3, even with the engine deeply throttled-down to 30%."** A reusable engine needs **33–100%** throttling to land at T/W near 1. dlr dlr	Conference	Medium	Confirms the throttle floor sets the minimum engine size.
Akin, UMd (S7)	Engine-out logic	Picked 6 engines so worst-case acceleration stays positive after losing one: (5/6)·1.3 = 1.083 > 1. umd	Academic	Medium	A simple way to model engine-out.
Krevor & Wilhite (2007)	—	Not retrieved or verified in this research	—	—	Don't cite until checked.

Landing T/W check (engineering calculation from published numbers): Assume a landing mass of 27–30 t, so weight is about 265–295 kN. One Block 5 engine at 57% gives 482 kN, or T/W ≈ 1.6–1.8. That is why Falcon 9 can't hover and does a single "hoverslam" burn. ycombinator At a fixed 7,605 kN total, the rated thrust per engine is 7,605/N kN. For T/W ⩽ 1 at minimum throttle, N ⩾ about 15–16 at 57% and about 11 at 40%. For DLR's T/W ⩽ 3, N ⩾ about 5. These limits cover the range of published answers (5–6 in Dresia, 9 on Falcon 9, 15–17 in DLR).

Discrepancies Between Sources

1. **Total sea-level thrust:** 7,605 kN (S1 text) vs 7,686 kN (S1 Table 2-1, S2) vs 7,607 kN (S5, S13). The cause is 845 vs 854 kN per engine. mit +2
2. **Liftoff mass:** 549,054 kg (official, S3) vs 569.3 t (S5). dlr spacex The 20 t gap means the S5 mass split doesn't match the official total.
3. **First-stage dry mass:** 27.2 t (S5) vs 25.6 t (S19) dlr wevolver vs community figures of about 22 t and up. None is official.

4. **Merlin 1D minimum throttle:** 57% (official, Block 5) vs 70% (2013 version, SpaceX statement quoted by NASASpaceFlight, S12) vs about 40% (forums).
5. **Merlin 1D chamber pressure:** 9.7 MPa (S13) vs 10.8 MPa (S19). Wikipedia wevolver
6. **Exit diameter:** 0.92 m (S13) vs about 1.07 m implied by 0.90 m² exit area (S19). Wikipedia wevolver
7. **Reusable LEO payload:** more than 17.4 t demonstrated (SpaceX on X, Starlink 5-2, Jan 2023; S16) vs 17.5 t or 18.5 t on different Wikipedia pages.
8. **Upper-stage vacuum thrust:** 934 kN (FT, S13) vs 981 kN (Block 5, official). Wikipedia +3
9. **H-1 mass:** 635 vs 911 vs about 1,000 kg. **RD-180 mass:** 5,393 vs 5,480 kg.

Recommendations: Baseline Simulation Inputs

Input	Baseline	Range for sensitivity	Basis
Total sea-level thrust (held constant)	7,605 kN	—	**Published** (S1)
Per-engine vacuum thrust ratio	914/845 = 1.082	1.07–1.10	Published estimate (S5), not official
Sea-level / vacuum Isp	283 / 312 s	±3 s	Published estimate (S5), not official
Minimum throttle	57%	40–70%	**Published** (S1), plus sensitivity
Engine T/W at 845 kN	184 (470 kg)	150–200	Secondary (S13, S12)
Engine mass vs thrust	Akin form scaled by 0.56 so it matches 470 kg at 845 kN	Exponent 0.8–1.0 on a power-law alternative	**Engineering assumption** calibrated to S7/S13
Thrust structure	2.55×10^{-4} kg/N × total thrust, plus a per-engine add-on of 0–50 kg	—	Published MER (S7) plus **assumption**
First-stage dry mass (9 engines)	26 t	22–27.2 t	Unofficial (S5, S19)
Recovery hardware	12% of dry mass	8–15%	Model proxy (S5, S6)
First-stage propellant	411 t (scaled so GLOW = 549 t)	400–419 t	Derived from S3 and S5, **assumption**
Landing propellant (drone ship)	25 t	15–35 t	Unofficial (S5); share check (S6)
Landing Δv	2,000 m/s	1,500–2,200 m/s	Published model value (S5)
Upper stage: thrust / Isp / dry / propellant / burn	981 kN / 348 s / 4.5 t / 111.5 t / 397 s	Isp ±3 s; dry ±0.5 t	Official thrust and burn time (S1, S3); others unofficial (S5)
Fairing + adapter	2.4 t, dropped at about 3 min	1.7–2.6 t	**Assumption** from S5 payload-bay value
Max-Q limit	35 kPa	30–40 kPa	**Unofficial proxy** (not published) orbitalradar
Ascent axial acceleration cap	4 g steady	3.4–6 g	**Assumption**, bounded by S14 telemetry (3.43 g) and S2 limit loads (6 g)
Landing T/W cap (single engine at minimum throttle)	⩽ 3	1–3	Published feasibility threshold (S17)

Input	Baseline	Range for sensitivity	Basis
Engine-out rule	Positive acceleration with N−1 engines	—	Published method (S7)
Launch site / target orbit	28.56° N (SLC-40); 200 km circular, 28.5°	Inclinations up to 55°	**Published** (S1, S2)
Validation target	22,800 kg expendable; more than 17,400 kg drone-ship reusable	±5%	Official (S3) / official (SpaceX on X, Starlink 5-2, S16)

What to do: Validate your model at N = 9 against the 22.8 t expendable and about 17.4 t reusable payloads. Then sweep N from 5 to 27 at fixed 7,605 kN, with the landing T/W cap and throttle floor as active constraints. Report the optimum as a band across the MER exponent and throttle floor, not as a single number. Based on the calculation above, the binding constraint at small N is likely landing T/W, and at large N it is per-engine fixed mass plus thrust-structure complexity.

Caveats

- No official SpaceX source gives Falcon 9 stage dry mass, propellant load, landing propellant, Isp, engine mass, max-Q, MECO state or touchdown velocity. Every such value here is an estimate and is labeled that way.
- The engine table in E1 came from search snippets collected by a research sub-agent. Primary pages (NASA MSFC F-1/H-1 fact sheets, Energomash engine pages) were not opened. Check them before quoting in the abstract.
- The Rohrschneider, Castellini, Tizón & Román and Krevor & Wilhite sources were not retrieved, so their functional forms are not given here.
- The landing T/W limits in Section F are engineering calculations from published inputs, not published results.
- Following this report's format, there is no separate bibliography at the end. The Source Key gives each source's title, author or organization, year and URL or DOI.